

CUSTOMER MILESTONE BRIEF

MS #50 - Oracle Semantic Coverage Program

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.50.0, 0.50.1, 0.50.2, 0.50.3
Date	2026-09-01
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #50 - Oracle Semantic Coverage Program converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Gives database architects a quantified, evidence-levelled Oracle coverage answer and advances it through broad core SQL, datatype, PL/SQL, transaction, and CDC execution tranches.

Why this matters to a customer

Gives database architects a quantified, evidence-levelled Oracle coverage answer and advances it through broad core SQL, datatype, PL/SQL, transaction, and CDC execution tranches.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- A catalog of 500 stable Oracle behavior contracts and 2,000 case specifications across ten exact domains.
- Bounded execution of 920 governed core SQL/type cases, 320 PL/SQL cases, and 280 transaction and operations cases.
- Transaction and operations coverage spanning commit, rollback, savepoints, isolation, locks, deadlocks, redo/LogMiner, metadata, session state, privileges, and diagnostics.
- Separately declared expectations and independently computed observations across 76 topic families, five behavior focuses, and four case dimensions.
- A deduplicated evidence ladder reporting 500 catalogued, 380 catalog-executed, 381 unique bounded-model verified, 0 native-verified, and 0 target-equivalent behaviors.
- Native execution plans plus deterministic artifacts, eight JSON Schemas, cross-platform verification, and adversarial dependency-drift, mutation, overlap, and overclaim tests.

How it differs from earlier work

MS #49 proved the mechanism with eight fixtures and 24 cases. MS #50.0 made them bootstrap evidence inside a governed broad-coverage program; MS #50.1 executes 920 core SQL/type cases, MS #50.2 executes 320 PL/SQL cases, and MS #50.3 executes 280 transaction and operations cases without promoting bounded-model evidence into native Oracle proof.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use oracle semantic coverage program as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- The 1,520 passed catalog cases are deterministic bounded-model evidence; they do not constitute native Oracle execution or conformance.
- The remaining 480 catalog cases have not executed.
- The 24 MS #49 bootstrap runs remain separate evidence records; seven behavior bindings overlap executed catalog tranches and the LOB binding remains bootstrap-only.
- Concurrency schedules, locks, redo/SCN, LogMiner, metadata visibility, privilege enforcement, and diagnostic observations are bounded simulations, not native platform evidence.
- No native Oracle 19c or 26ai execution receipt exists.
- Catalogued is not supported; bounded-model verified is not native conformant; native verified is not target equivalent.
- iDempiere projection, CloudBank mapping, application equivalence, migration completion, and production readiness remain false.

Source of record

- CHANGELOG.md release section(s): 0.50.0, 0.50.1, 0.50.2, 0.50.3
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity